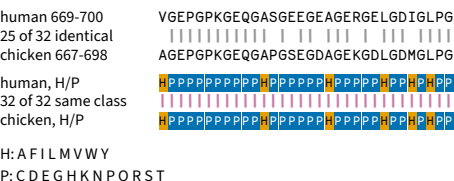


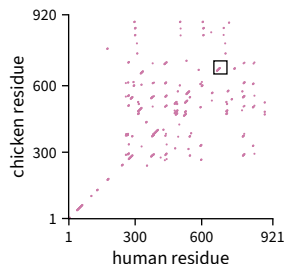
H: hydrophobic class
 P: polar class
 — H/P alphabet
 — 20 amino acids (no reduction)
 - - - k^* : H/P k-mer length expected once by chance in Swiss-Prot

a How kmerseek finds a shared region: human vs chicken COL9A1, hp_thomas_dill_no_c2, $k = 19$

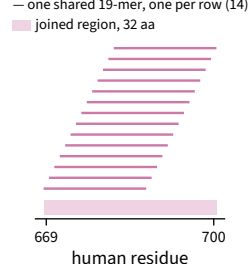
1 Encode each residue as H or P



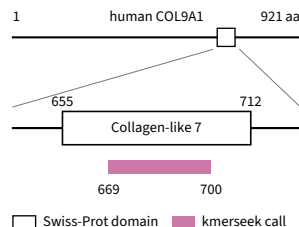
2 Seed: shared 19-mers



3 Join overlapping 19-mers

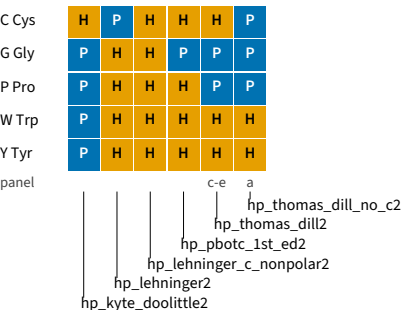


4 Compare with Swiss-Prot

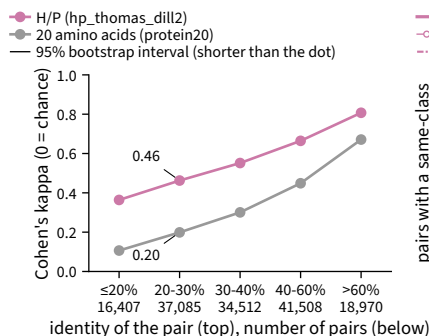


b Six H/P alphabets differ on five residues

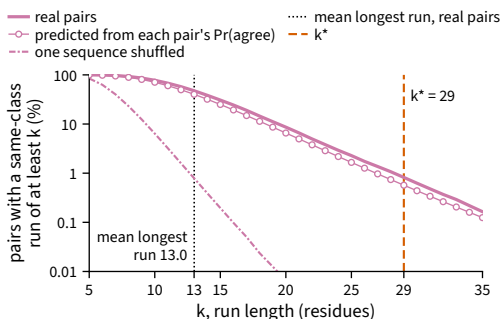
H in all six: A F I L M V
P in all six: D E H K N Q R S T



c Kept above chance at every identity



d but in runs of about 13 residues



e A typical pair at 20-30% identity: its longest same-class run, 12, is the median of all 37,085

Pfam HTH_6 (PF01418), RPIR_ECOLI and Y143_HAEIN, hp_thomas_dill2

